Historical Tire Accuracy & Evidence Report

Maserati 250F 6 cylinder

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OVERALL ASSESSMENT

Historical evidence confidence: 77 / 100

Verdict: Good evidence-weighted historical reconstruction

This score describes evidence completeness and confidence in the selected historical tire context.
It is not a probability that proprietary factory coefficients are exact.

1. Historical identity and context

Car: Maserati 250F 6 cylinder [Direct AC package]
Racing year: 1957 [Historical research / inferred]
Series / class: 1954-58 Formula 1 / Grand Prix [Historical research / inferred]
Tire supplier: Pirelli [Historical research / inferred]
Construction: bias [Selected historical/tire-family context]

2. Current generated tire model

Historical tire category: FAM002 - 1950s GP / sports treaded bias
Class calibration: CLS102 - 1954-58 Formula 1 / Grand Prix
Compounds: Period Treaded Race
Vehicle mass used: 700 kg
Static front weight used: 51.0%
Front tire geometry: 96 mm width, 668 mm OD, rim radius 203.2 mm
Rear tire geometry: 124 mm width, 717 mm OD, rim radius 203.2 mm
Vertical rate: 49.9 N/mm front / 64.5 N/mm rear
Reference load FZ0: 1107 N front / 1385 N rear
Thermal load duty: F medium 11.53 N/mm | R medium 11.17 N/mm; transparent N/mm reconstruction thresholds
Compound/load checklist: Period Treaded Race [pass]
Ideal hot pressure: 32.0 psi front / 38.0 psi rear
Pressure generation: Auto-solved using AMBIENT_PROXY_UNRESOLVED initial core; ambient 26.0 deg C retained separately
Medium generated cold pressure: 27.3 psi front / 32.9 psi rear
Imported AC cold-pressure reference: None / not imported
Blanket temperature: 70 deg C
Thermal implementation: CSP Thermal Model V2 + required Kunos thermal sections
Physics output: CSP Extended Physics - Thermal V2
Car physics requirement: car.ini [HEADER] VERSION=extended-2
Extended contact rays: 2 lateral / 4 longitudinal per side, 60 deg
Knowledge release: v1.7.1 / schema 1.2.0
Historical family physics: FAM002: family-specific grip/load/transient/thermal priors active

Imported old tire reference: RATE 283800 F / 273800 R N/m; ideal pressure 40 psi (reference only)
Period Treaded Race front Thermal V2 reconstruction: mass 5.35 kg; volume 22.85 L; FRICTION_K 0.013924; CARCASS_ROLLING_K 0.285351; SURFACE_TO_AMBIENT 0.077082; confidence 77%
Period Treaded Race rear Thermal V2 reconstruction: mass 6.88 kg; volume 36.64 L; FRICTION_K 0.013798; CARCASS_ROLLING_K 0.286149; SURFACE_TO_AMBIENT 0.075426; confidence 77%
Validation track: Manual circuit / energy 1.00

3. Historical-accuracy findings

- Car identity used for this tire pack: Maserati 250F 6 cylinder.
- Selected historical racing context: 1954-58 Formula 1 / Grand Prix.
- Selected supplier context: Pirelli.
- Historical tire category FAM002 (1950s GP / sports treaded bias) is active; it controls construction/type, period menu and class-life wear priors without inventing supplier-proprietary peak grip.
- Tire construction (bias/cross-ply) was inferred from the researched historical tire designation.
- Wear-to-grip behavior remains provisional/evidence-weighted rather than historically certified from period stint data.
- Terminal tire failure is a simulation layer: 50% grip after +0.20 vKm beyond the final normal-wear point. It is not direct historical puncture evidence.
- Generated cold pressures use an explicit AC initial-core state (26.0 deg C, AMBIENT_PROXY_UNRESOLVED); ambient 26.0 deg C is retained separately and is not silently equated with tire core.

4. Wear and degradation status

Wear calibration status: provisional
Wear calibration note: ACLM historical class-life prior: Soft 60 km, Medium 90 km, Hard 190 km, Intermediate 75 km, Wet 90 km. Wear shape remains evidence-weighted and should be telemetry-certified for the specific car/track.
Terminal failure: Enabled: 50% grip after +0.20 vKm
Wear philosophy: 100% AC wear represents the best historical baseline; users can use AC wear multipliers to compress race strategy.

5. Assetto Corsa implementation status

- Assetto Corsa tire format VERSION=10.
- Front/rear tire sections include load curves, pressure model, FZ0, relaxation length, camber LUTs and combined-slip parameters.
- Every selected compound exports its front/rear wear LUT and temperature/performance LUT.
- Every pack exports ACLM_LOAD_COMPOUND_CHECKLIST.txt with series-menu and axle-duty assessments.
- The pack validator blocks export for missing referenced LUTs, malformed LUT rows, invalid geometry or other structural errors.
- This PDF is generated automatically and bundled with every Tire Lab ZIP export.

6. Evidence provenance

Direct AC-package data describes what the imported mod currently contains; it is not automatically historical proof. Researched context is public historical evidence. Reconstructed values are Tire Lab engineering/calibration choices.

- Imported AC package files inspected for generation: ui_car.json, car.ini, tyres.ini, suspensions.ini, setup.ini.
- Green Tire Lab fields are direct values from the imported AC package. These document the mod state, not necessarily historical truth.
- Amber fields are historical research/inference and are retained as reviewable evidence rather than hidden assumptions.
- Unmarked values are user selections, defaults or Tire Lab reconstruction/calibration values.
- Thermal V2 coefficients are physical reconstruction priors derived from geometry, load, rate, rolling resistance, construction, pressure, drivetrain/brake duty and class evidence; they are not confidential supplier data.
- CSP parameter meanings and obsolete-key handling follow the official CSP Tyre Thermal Models V1/V2 documentation.

7. Sources

No public historical source pages were attached to this export. Run Historical Research in Tire Lab before export if you want researched class/supplier sources embedded in the report.

8. Known limitations / next validation

- A generated AC tire can be structurally valid while exact supplier proprietary coefficients remain unavailable.
- Peak friction, exact stiffness, thermal constants and wear-to-grip curves should only be described as factory-exact when primary evidence exists.
- Lap time and Assetto Corsa behavior are validation evidence for the simulation implementation, not standalone proof of historical tire specifications.
- User-adjustable AC tire-wear multipliers can compress or extend race strategy without changing the historical baseline wear curve.
- Light/medium/heavy thermal load-duty thresholds require telemetry validation for each car, track, setup and ambient condition.